Homework 1 (Due: 8th Oct.)

(1) Do the following signals require the same number of sampling points as $x(t)$? Why? (a) $x(t-2)$; (b) $x^2(t)$; (c) $x(2t)$; (d) $2x(-t)$ (15 scores)

(2) Which of the following applications are the proper applications of the short-time Fourier transform? Also illustrate the reasons. (a) Signal sampling. (b) convolution computation; (c) image identification; (d) animal voice analysis. (15 scores)

(3) (a) Why the sinc function may not reflect the frequency distribution of a rectangular function? (b) Suppose that

$$x(t) = 1 \text{ for } -1 < t < 1, \quad x(t) = 0 \text{ otherwise.}$$

Determine the rec-STFT of $x(t)$ if $B = 1$. (10 scores)

(4) (a) How does the parameter σ affect the resolution of the scaled Gabor transform ?

(b) If we want to analyze a vocal signal (the units in the t-axis and the f-axis are second and Hz), should we use a larger or a smaller value of σ? Why? (15 scores)

(5) (a) What is the relation between a rectangular function and a Gaussian function? (b) Why better time-frequency analysis result can be obtained if one uses the Gaussian window instead of the rectangular window? (10 scores)

(6) Determine the scaled Gabor transform of the following functions.

(a) $\delta(t)$, (b) 1, (c) $\exp(-\pi t^2)$. (15 scores)

(7) Write a Matlab or Python program that can generate a *.wav file whose instantaneous frequency is $\pm(at^2 + bt + c)$ Hz, the length of the file is T second, and the sampling frequency is Fs Hz.

gwave (a, b, c, T, Fs)

The code should be handed out by NTUCool. (20 scores)

(Extra): Answer the questions according to your student ID number.

(ended with 0, 1, 2, 3, 5, 6, 7, 8)